Babu Reddy S

Bangalore, India

+91 8431017886 | babureddys003@gmail.com | linkedin.com/in/babure | github.com/babure

SUMMARY

Software Engineer with 4 years of experience building scalable backend systems and AI-driven platforms in Java (Spring Boot) and Python (FastAPI), with React.js on the front end. Applied GenAI, RAG, SLM fine-tuning, and classical ML (classification, clustering, anomaly detection) across a real-time voice AI platform, an LLM-agent test automation platform, and an AI-assisted monitoring platform. Works across product, QA, DevOps, and customer stakeholders to ship production systems.

TECHNICAL SKILLS

Languages	Java, Python, JavaScript, SQL, HTML, CSS
Java Ecosystem	Spring Boot, Spring Data JPA, Spring Security, Hibernate, JPA, Maven, JUnit, Mockito
Python Ecosystem	FastAPI, SQLAlchemy, Alembic, Pydantic, asyncio, Uvicorn, Pytest
Frontend	React.js, Redux, Jest, React Testing Library
AI/ML & LLM	Generative AI, LLM Integration, RAG (Retrieval-Augmented Generation), SLM Fine-Tuning, AI Agents, Prompt Engineering, Structured Outputs, ML Classification, Clustering, Anomaly Detection, MCP (Model Context Protocol), Real-Time Voice AI, Speech-to-Text, Text-to-Speech, LiveKit, Deepgram, OpenRouter
AI-Native Dev Tools	Cursor, Claude, OpenAI Models
APIs & Security	REST APIs, OpenAPI/Swagger, JWT, OAuth2, Role-Based Access Control
Databases & Caching	PostgreSQL, MySQL, MongoDB, Redis
DevOps & Cloud	Git, GitHub, Docker, Kubernetes, Jenkins, AWS (EC2, S3, Lambda, CloudWatch), Caddy, Postman
Testing & Automation	API Testing, Test Automation, Low-Code/No-Code Test Automation, Workflow Automation
Methodologies	Agile, Scrum, JIRA

PROFESSIONAL EXPERIENCE

Software Engineer — Oct. 2022 – Present

Avekshaa Technologies, Bangalore, India

- Partnered with 10+ product managers, QA, and DevOps engineers and business stakeholders in an Agile/Scrum environment to turn requirements into designs, align on priorities and release plans, and own 3 production platforms from design through deployment.

Project: Zenovia (AI Mock Interview & Candidate Screening Platform) — *Jun. 2026 – Present*

AI mock interview and candidate screening platform with live voice interviews, ATS-style scoring, and automated reports.

- Architected an async FastAPI backend (asyncio, Pydantic, SQLAlchemy/Alembic) integrating 4 third-party services (LiveKit, Deepgram, OpenRouter, Razorpay) with timeouts, retries, and fallbacks, serving ∼50K requests per day at p95 latency under ∼300ms; tuned PostgreSQL with indexing, query optimization, and connection pooling, and added Redis caching with a ∼70% hit rate, cutting average query time by ∼50%.
- Designed stateless, Docker-containerized services behind Caddy that scale horizontally to 25–50 concurrent live interview sessions, and moved heavy work (ATS scoring, report generation) to background jobs so reports are ready within ∼30s of an interview ending; handled Razorpay payment events through verified, retry-safe webhooks across 4+ plan tiers with zero duplicate charges over 1,000+ events.
- Collaborated with product, QA, and DevOps teams and ran 10+ feedback sessions with university and company users to gather requirements, prioritize 20+ features through weekly demos, and align on deployment and integrations, taking the platform from concept to production.
- Engineered a real-time voice interview pipeline (speech-to-text, LLM reasoning, text-to-speech) on LiveKit with Deepgram STT/TTS for an AI interviewer, keeping per-turn response latency under ∼1.5s and reaching ∼99% session completion, with automatic fallback to LiveKit's built-in STT/TTS if a provider is unavailable.
- Developed the LLM layer on OpenRouter for adaptive question generation, response evaluation, and ATS-style report generation, using prompt engineering, Pydantic-validated structured outputs (∼98% valid on first pass), and model routing by task to score candidates on 8 evaluation dimensions while cutting LLM cost per interview by ∼35%.

- Leveraged MCP (Model Context Protocol) tools with Cursor and Claude to bring vendor docs into the dev workflow, cutting integration research time by ∼30%.

Project: HuLA (Human-Like Automation Platform) *Apr. 2025 – May 2026*

Low-code/no-code test automation platform where an AI agent generates test scripts from plain-English prompts.

- Architected the Spring Boot backend (20+ REST endpoints, OpenAPI/Swagger) to serve ∼2M requests per month, reducing p95 API latency from ∼800ms to ∼200ms (∼75%) through Redis caching, pagination, and JPA/Hibernate query tuning; split data across PostgreSQL (metadata) and MongoDB (execution results) with indexing, cutting dashboard query time by ∼60% as run history passed 1M records.
- Orchestrated a queue-based distributed execution engine that dispatches test runs to containerized browser workers on Kubernetes, supporting 100+ parallel executions and 1,000+ test runs per day, and cutting QA regression cycles from ∼1 day to under ∼1 hour.
- Launched a multi-tenant, no-code product used by 10+ QA teams, with project-level isolation, JWT-based authentication and RBAC across 3 user roles, and 15+ React.js screens including live execution dashboards for projects, scenarios, test cases, and workloads.
- Pioneered an LLM-powered agent workflow that explores a target website, understands page structure, and converts a plain-English test description into an executable test script, reducing manual test authoring from roughly 2 hours to about 15 minutes per scenario (∼85%).
- Integrated a Retrieval-Augmented Generation (RAG) pipeline over 200+ collected functional and business requirement documents (FRD/BRD), grounding test-case and script generation in project requirements, improving requirement coverage by ∼40% and cutting manual edits to generated tests by ∼30%.
- Trained a domain-specific small language model (SLM) on 5,000+ FRD/BRD and test-script examples to generate test scripts, with ∼90% of outputs executing without edits and inference cost and latency lowered by ∼60% and ∼50% versus large hosted models.

Project: DEM (Digital Experience Monitoring Platform) *Oct. 2022 – Mar. 2025*

Dynatrace-like monitoring of web/mobile apps, servers, CI/CD pipelines, and infrastructure with AI-assisted root cause analysis.

- Established a high-throughput ingestion pipeline that collects metrics from browser and server agents across ∼50 applications and 50+ servers, handling ∼50M events per day with batching and queue-based buffering so traffic spikes never dropped data, and delivering metrics to dashboards within ∼5 seconds.
- Optimized the metrics data layer with partitioning, indexing, downsampling, and tiered retention (raw data 7 days, rollups 13 months), cutting storage cost by ∼50% and keeping dashboard queries under ∼2 seconds across billions of stored data points.
- Deployed a real-time monitoring product with distributed synthetic monitoring (30+ business transactions every 30 minutes, ∼1,440 runs per day) that detected outages ∼10 minutes before user reports, lightweight browser agents collecting 10+ frontend metrics at under ∼1% page overhead, CI/CD pipeline monitoring, and deduplicated Webhook and Slack alerting.
- Introduced a GenAI incident-description feature that turns raw logs, metrics, and alert context into human-readable summaries for ∼90% of incidents, cutting on-call triage time by ∼35%.
- Applied ML classification (∼90% accuracy across 8 incident categories) and clustering to group related alerts, reducing alert volume reaching engineers by ∼50%.
- Implemented anomaly detection on application and infrastructure metrics (∼85% precision, false positives down ∼40%) and AI-assisted root cause analysis that correlates logs and metrics across the stack, reducing mean-time-to-resolution (MTTR) by ∼40%.

EDUCATION

B.E. in Computer Science and Engineering Jul. 2018 – Jul. 2022

Nagarjuna College of Engineering and Technology, Bangalore, India

ACHIEVEMENTS

- Best Performer at Avekshaa 2024, recognized for technical contributions and reliability improvements across core platform services.

LANGUAGES

- English • Hindi • Kannada • Telugu